

 FINANCIAL RISK & LOSS ANALYTICS

Executive Fraud Risk & Loss Mitigation Strategy

End-to-End Analytics & Predictive Insights on \$1.13M Fraud Exposure



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Executive Summary: Comprehensive end-to-end data pipeline analyzing high-volume transaction channels, peak vulnerability windows, and targeted automated controls to curtail financial loss while maintaining optimal checkout speed.

Baseline Risk Metrics

TOTAL FRAUD LOSS

\$1.13M

TOTAL FRAUD CASES

2,145

OVERALL FRAUD LOSS RATE

2.94%

AVG FRAUD TRANSACTION

\$528.35



Primary Vulnerability Identified: Online channels drive **63.4% (\$718.35K)** of overall financial loss exposure compared to In-Person (POS) transactions. High-risk digital checkouts represent the primary channel gap requiring immediate rule intervention.

🕒 Attack Vectors & Peak Windows

🛒 Category Exposure Breakdown

Shopping (Net) and Online Services account for over 65% of unauthorized digital transactions, led heavily by card testing scripts.

🌙 Peak Time Windows

Fraud activity peaks sharply during late-night and early-morning hours (10:00 PM – 3:00 AM), taking advantage of lower manual review coverage.

🔒 Key Security Gap

Unrestricted low-friction checkout protocols during off-peak night hours create a key structural gap for automated testing scripts.



Actionable Mitigation Rules



Rule 1: Dynamic Multi-Factor Authentication

Automated step-up MFA challenge triggered for high-risk online transactions > \$500 initiated during off-peak hours (10:00 PM – 5:00 AM).



Rule 2: Velocity & Capping Controls

Automated temporary holds triggered when a card exhibits >3 consecutive high-value online attempts within a 15-minute rolling window.



Projected Annual Financial Impact

Targeted friction blocks top 20% high-risk night spikes with minimal impact to legitimate shoppers.

\$180K – \$220K
~15%–20% Loss
Reduction

🔧 Technical Architecture & Pipeline

</> Data Engineering (Python & SQL)

Automated ingestion of card transaction streams, missing value imputation, and feature engineering for time/channel attributes.

🗄️ Data Modeling (Power BI DAX)

Optimized Star Schema design centered around a main Fact table and dedicated DAX measure repository for loss rates and KPI trends.

📊 BI & Interactive Dashboard

Executive Power BI dashboard enabling real-time drill-downs, state slicing, and hourly trend analysis for risk teams.

🔗 Technical Artifacts
Ready

📄 GitHub Repository & Interactive Portfolio Demo
Available

End-to-End data pipeline workflow for Financial Risk Dashboard

Stage 1: ETL & Data Engineering (Python + SQL)



Stage 2: Business Intelligence (Power BI)

